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Research Paper

Volume and location of screen-detected lung nodules associated with lung cancer within two-year follow-up: Post hoc analysis from the UK Lung Cancer Screening (UKLS) trial

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ABSTRACT

Introduction: Follow-up of radiologically suspicious lung nodules from screening programs remains complex and unstandardized. This study analyses distribution and characteristics of solid and part-solid non-calcified nodules ≥100 mm^3 and non-solid nodules ≥8 mm detected at baseline in the UK Lung cancer Screening (UKLS) trial, with insights into their relationship to screen-detected lung cancers.

Methods: UKLS participants with baseline indeterminate and positive lung nodules were included. Nodule location was categorized by lobe and by attachment: intraparenchymal, juxta pleural or pleural-based. Solid and part-solid were classified based on semi-automated volume: 100–300 mm^3 and ≥300 mm^3, and non-solid nodules based on diameter ≥8 mm. This measurement approach aligns with UKLS protocol and the British Thoracic Society (BTS) guideline. Their relationship to screen-detected lung cancer was based on histological outcomes after a 2-year follow-up period.

Results: 279 UKLS participants with 373 solid or part-solid lung nodules ≥100 mm^3 and 57 participants with 62 nonsolid nodules ≥8 mm were included. Among solid and part-solid nodules, 233 were 100–300 mm^3, with 39.5% in upper lobes and 72.5% intraparenchymal; 140 nodules were ≥300 mm^3, with 51.4% in upper lobes and 53.6% intraparenchymal. Nodules ≥300 mm^3 group were more likely to be in the upper lobes (OR 1.97 [95% CI 1.12–3.48]). Within 2 years, 34 solid or part-solid nodules were diagnosed as lung cancer, yielding a 2-year cancer probability of 9.1% (95% CI 6.6–12.5). The 2-year lung cancer probability was higher in nodules ≥300 mm^3 (RR 18.1, 95% CI 5.6–58.4; $p < 0.001$) and in upper lobe nodules (RR 2.7, 95% CI 1.3–5.5; $p = 0.009$).

Conclusion: Solid and part-solid nodules with volume ≥300 mm^3 were more often found in the upper lobes and were associated with a higher probability of lung cancer within a two-year follow-up period.

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1. Introduction

Lung cancer has the highest incidence and mortality rates among all cancers, representing 12.4% of new global cancer cases and 18.7% of cancer-related deaths. In Europe, there were 484,306 new cases and 375,569 deaths in 2022, equivalent to one death every 83 s and one new case every minute [1,2]. Although recent treatments for lung cancer like surgery, radiotherapy, targeted therapy, and immunotherapy have advanced significantly showing promising results to prolong patient survival, one of the most effective methods to reduce lung-cancer deaths is via early detection and optimal nodule management in low-dose computed tomography (LDCT) lung cancer screening (LCS) programs [3–5]. Early lung cancer diagnosis significantly improves patient outcomes and survival rates. Field et al. performed a meta-analysis of nine LCS trials, showing a significant reduction in lung cancer mortality with a pooled overall relative rate of 0.84 (95% confidence interval [CI] 0.76–0.92) [3]. Furthermore, LDCT LCS has been proven cost-effective in high-risk individuals [6]. Hence, many countries are implementing national screening programs.

Widespread implementation of LCS requires a robust lung nodule management strategy. Lung nodule management strategies must yield not only accurate results but should also be suitable for easy adoption in clinical practice. Multiple nodule management protocols are available, all of which categorise lung nodules based on nodule subtype (solid, part-solid, or non-solid) and size [7–9]. To date, nodule location or other nodule characteristics such as attachment have not played a prominent role.

The highest nodule category determines individual follow-up strategy: negative cases go to the next screening round, indeterminate cases receive a short-term follow-up to assess nodule growth, and positive cases are immediately referred for further work-up [10]. Indeterminate lung nodules are often defined as suspicious nodules with a volume of 100–300 mm^3 or a diameter of 5–10 mm [10,11]. Positive nodules, defined as nodules with a volume $\geq$300 mm^3, are referred immediately, as the probability of lung cancer is high at 16.9% (95% CI 14.1–20.0) in this subgroup [11]. However, once referred, further work-up strategy for these nodules is not standardized. Pulmonologists, oncologists, and thoracic surgeons often consider nodule characteristics including location, attachment, and size when deciding on the next diagnostic steps, yet there remain very limited studies describing location distribution of nodules found in the lung cancer screening programs. In the Dutch-Belgian Lung Cancer Screening (NELSON) trial, Horeweg et al reported 45% of all lung cancers were located in the right upper lobe, indicating nodules in this location may require more rigorous work-up [12]. Nevertheless, this hypothesis is yet to be validated in a different lung cancer screening population.

Evidence on the association between nodule characteristics such as size and location, and ultimate lung cancer diagnosis in a LDCT screening population is needed for shaping and standardising nodule management guidelines in the future. Therefore, this study aims to systematically evaluate the distribution and characteristics of indeterminate (100–300 mm^3) and positive ($\geq$300 mm^3) solid or part-solid nodules, as well as non-solid nodules ($\geq$8 mm), detected at baseline in participants of the UK Lung Cancer Screening (UKLS) trial. Nodule measurement was performed in accordance with the UKLS protocol for nodule classification and management and the British Thoracic Society (BTS) guidelines, with volumetric assessment applied where feasible and appropriate for solid nodules and the solid part of part-solid nodules. In contrast, pure non-solid nodules were assessed using diameter-based criteria due to limitations in reliable volumetric segmentation caused by indistinct margins and variable attenuation, with diameter thresholds remaining guideline-recommended (8). This study further investigates the association between these nodule characteristics and subsequent histologically confirmed lung cancer diagnosed during the two-year follow-up period.

2. Methods

2.1. Study design and population

This retrospective study used available data on all non-calcified solid and part-solid lung nodules with a volume $\geq$100 mm^3 and nonsolid nodules with diameter $\geq$8 mm detected at baseline in UKLS trial participants. Screen-detected solid and part-solid nodules were further classified by volumetric size into indeterminate (100–300 mm^3) and positive ($\geq$300 mm^3). Assuming a spherical nodule geometry, these volumetric thresholds correspond to approximate maximum diameters of 6 mm for 100 mm^3 and 8 mm for 300 mm^3. The UKLS 'Wald Single-Screen' LDCT randomised-controlled trial took place in Liverpool Heart & Chest Hospital (LHCH), Merseyside and Papworth Hospital, Cambridgeshire, United Kingdom (registration number: ISRCTN78513845). A total of 4055 high-risk individuals were randomised (1:1) to either the LDCT screening or control arm. High-risk individuals were identified using Liverpool Lung Project risk model version 2 ($LLP_{version}2$) with risk score 4.5% or above. Details on inclusion and exclusion criteria of the UKLS trial were published previously [3].

2.2. LDCT thorax scan protocol

Participants in the screening arm received a baseline LDCT scan with the following specifications: 16 + channel multidetector CT, no contrast,100–140 kVp, single breath hold at maximum inspiration craniocaudally from lung apices to bases. LDCT scans were acquired using a Siemens Definition Flash 128 slice scanner (Siemens, Erlangen, Germany) at LHCH and Papworth Hospital. During the initial phase of the study, in 2011, a Philips (Best, The Netherlands) Brilliance 65 scanner was utilized at LHCH. Images were reconstructed at 1 mm thickness and 0.7 mm increment.

2.3. Nodule detection, measurement, and classification

During the UKLS trial, two thoracic radiologists examined all baseline LDCT scans, and a third senior thoracic radiologist reviewed any discrepancies and determined the final nodule classification. All scans were read using LungCare (LungCare, version Somaris/5 VB 10A. Siemens Medical Solutions) on Syngo Siemens workstations. No computer-aided detection was used; however, semi-automated volume segmentation was employed for lung nodule measurements. For part-solid nodules, volumetric measurement represented the solid component of the nodule and was performed using semi-automated software, whereas non-solid nodules were measured manually using calipers to obtain diameter. Volumetry was applied where feasible and is recommended as the preferred measurement method for solid and part-solid nodules. In contrast, volumetric segmentation of non-solid nodules is unreliable due to indistinct margins and variable attenuation; therefore, diameter-based thresholds remain the standard for non-solid nodules in NELSON trial, the BTS guidelines, and Lung-RADS, including the UKLS protocol (3, 5, 8, 10). Detected nodules were classified during the trial into four categories according to the same protocol used during the NELSON trial [5] (see Table S1). For this study an updated version of the NELSON study protocol, NELSON 2.0, was used [13], which is more in line with recent nodule management guidelines [9,14].

2.4. Indeterminate and positive lung nodules analysis

This study focused on participants with indeterminate and positive solid or part-solid lung nodules measuring $\geq$100 mm^3 in volume (see Fig. 1), or $\geq$8 mm in diameter in the case of non-solid nodules. Nodules demonstrating benign characteristics, either intra-pulmonary lymph nodes or a benign calcification pattern, were excluded. Included solid or part-solid nodules were divided into two groups based on volumetric size: indeterminate nodules with a volume of 100–300 mm^3 and positive

nodules with a volume of ≥300 mm^3. Nodule location (by lobe), attachment type (intraparenchymal, juxtapleural, and pleural-based), and nodule density were analysed both overall and according to size category.

2.5. Histologically confirmed lung cancer analysis

All histologically confirmed lung cancers detected, whether baseline screen-detected or diagnosed within the 2-year follow-up period after the baseline LDCT scan, were match to the corresponding lung nodule at baseline. Characteristics of nodules matched to histologically confirmed lung cancer were analysed based on nodule volume, location, density, lung cancer stage and histological subtype.

2.6. Statistical analyses

Descriptive analyses were conducted to summarize the characteristics of participants, lung nodules, and lung cancers. Participant characteristics included patient demographics, smoking status, asbestos exposure, self-reported history of respiratory disease, history of solid cancer and $LLP_{version2}$ risk score. Nodule characteristics included volume, lobar location, attachment type, and density, stratified by indeterminate and positive size categories. Lung cancer characteristics included the same variables as nodules, with the addition of TNM stage version 8 and histology, stratified by time of diagnosis; at baseline or within 2 years after baseline. The lobar distribution of lung nodules (by size category) and histologically confirmed lung cancers were visualized using coronal slice of the lung.

To evaluate statistical differences between the characteristic of the nodules sized 100–300 mm^3 and ≥300 mm^3, we accounted for within-participant clustering by including a random intercept. Participant clustering indicates that some nodules could have originated from the same patients. For continuous variables, data distribution was first assessed. If normally distributed, a linear mixed-effects model was applied. Otherwise, a non-parametric clustered Wilcoxon rank-sum test was used. For categorical variables, mixed-effects logistic regression models were utilized. Binary outcomes included lobe location (upper vs other), nodule density (solid vs part-solid), and attachment type (intraparenchymal vs non-intraparenchymal), with volume category as the main predictor. Odds ratios (OR) with 95% confidence intervals (95% CI) were calculated to compare the odds of each outcome between nodule volume categories.

The two-year probability of lung cancer in participants with at least an indeterminate baseline nodule was estimated based on nodule volume, lobar location, attachment type, and density. For each variable, malignancy probability was calculated as the proportion of malignant nodules among the total number of nodules in that category. Confidence intervals for the proportions were computed using the Agresti–Coull method. Differences in proportions between two groups within each category were assessed using mixed-effects logistic regression models accounting for within-participant clustering. Risk estimates with 95% CI were reported to describe differences between groups. Statistical significance was defined as $p < 0.05$. All analyses were conducted using SPSS version 29 and R version 4.5.0.

3. Results

3.1. Participant characteristics

Of the 1994 participants who received baseline CT, lung nodule data

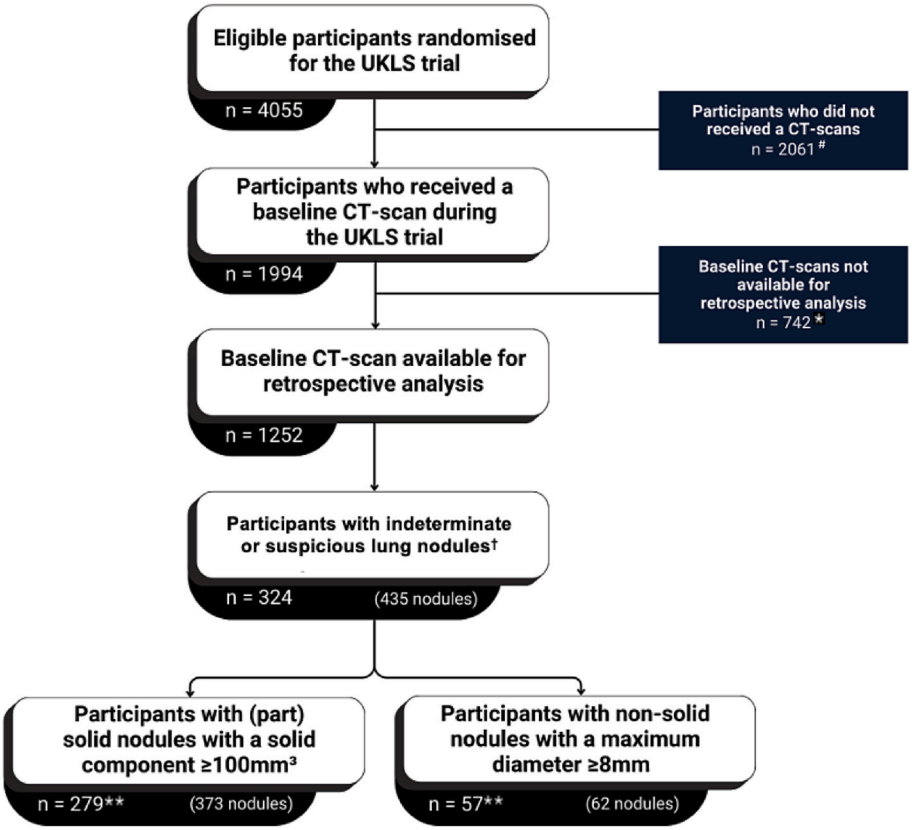


Fig. 1. Participant Flowchart. #2027 participants were in the control arm and 34 participants were in the screening arm but did not receive baseline scan †Nodules not considered to be benign due to either an intrapulmonary lymph node appearance or benign calcification pattern. *The majority of unavailable nodules (n = 666) were classified as negative with no significant indeterminate nodules during the UKLS trial **12 participants had both solid or part-solid nodules ≥100 mm^3 and non-solid nodules ≥8 mm.

from 1252 participants were available for retrospective analysis in this study (see Fig. 1), the large majority of those unavailable (n = 666) being classed as negative (no significant indeterminate nodule) during the UKLS trial. At baseline CT screening, 324 participants had 435 radiologically suspicious nodules with either a volume ≥100 mm^3 in the case of solid or part-solid nodules or diameter ≥8 mm for non-solid nodules. Of these, 362 solid and 11 part-solid nodules were detected in 279 participants, and 62 non-solid nodules in 57 participants, with 12 participants having nodules in both nodule density categories (see Fig. 1 and S1, Table S2 and S3). Included participants had a median age of 68.9 years (IQR 65.9–72.3), 22.8% were female, 98.5% white, 39.2% individuals who currently smoke, and 60.5% individuals who formerly smoked. Former smokers had quit smoking for a median of 14 years (IQR 8–26), and the median smoking duration for all participants was 45.5 years (IQR 35.3–51.0). Median $LLP_{version}2$ risk score was 10.5 (IQR 7.2–16.5) and 42% of all participants had a prior history of solid malignant tumors (excluding lung cancer) (see Table 1, S2 and S4). Baseline demographic characteristics of the overall UKLS cohort are provided in Table S5 as a reference.

3.2. Indeterminate and positive solid, part-solid and non-solid lung nodules

Of 373 radiologically suspicious solid or part-solid nodules, 233 (62.5%) had a volume 100–300 mm^3. Median volume in this group was 145.9 mm^3 (IQR 114.6–194.5 mm^3), with 39.5% located in the upper lobes and 72.5% located intraparenchymally. In the ≥300 mm^3 group, 140 nodules (37.5%) were identified with a median volume of 709.7 mm^3 (IQR 422.0–1669.2 mm^3), with 51.4% located in the upper lobes. Of these, 53.6% were intraparenchymal and 32.9% pleural based. Nodules in the ≥300 mm^3 group were more likely to be located in the upper lobes with OR 1.97 (95% CI 1.12–3.48) as compared to nodules sized 100–300 mm^3. No difference in density was observed between the two groups (see Table 2 and Fig. 2). A total of 62 non-solid nodules were identified, with a median diameter of 9.6 mm (IQR 7.9–12.1). Most were located intraparenchymally (80.6%) and were located in the right upper lobe (RUL 30.6%), left upper lobe (LUL 29.0%), and right lower lobe (RLL 27.4%) (see Table S3).

Table 1
Characteristics of participants with an indeterminate or positive lung nodule.

Participant characteristics	n = 324
Age (years), median (IQR)	68.9 (65.9–72.3)
Female sex, n (%)	74 (22.8)
Ethnicity group	
White	319 (98.5)
Mixed Asian	1
Mixed Black	1
Unknown	3
Current smoker, n (%)	127 (39.2)
Former smoker, n (%)	196 (60.5)
Years since cessation of smoking, median (IQR)	14.0 (8.0–26.0)
All smoking duration in years, median (IQR)	45.5 (35.3–51.0)
Asbestos exposure, n (%)	135 (41.7)
Self-reported history of respiratory disease, n (%)	
Tuberculosis	21 (6.5)
Pneumonia	61 (18.8)
Emphysema	22 (6.8)
COPD	70 (21.6)
Asthma	73 (22.5)
Bronchitis	266 (82.1)
Previous history of cancer*, n (%)	136 (42.0)
Family history of lung cancer, n (%)	
Yes < 60 years	112 (34.6)
Yes ≥60 years	105 (32.4)
$LLP_{version2}$ risk score, median (IQR)	10.5 (IQR 7.2–16.5)

IQR = inter quartile range; COPD = chronic obstructive pulmonary disease; LLP = Liverpool Lung Project.
* Excluding non-melanoma skin cancer.

3.3. Histologically confirmed lung cancer; screen-detected or diagnosed within 2 years follow-up

During a follow-up period of up to 2 years after the baseline CT scan, lung cancer was diagnosed in 34 participants (34 nodules), all arising from solid or part-solid nodules. No cancers were detected in nonsolid nodules. The majority of lung cancers (91.2%) occurred in those with a baseline volume ≥300 mm^3. Demographic and clinical characteristics of lung cancer participants are presented in Table S4. Participants with lung cancer were significantly more likely to currently smoke (p = 0.043), have a longer smoking duration (p = 0.003), a history of solid tumors (p = 0.018), and a higher $LLP_{version}2$ risk score (p < 0.001) compared to those without lung cancer within 2 years after baseline.

Not all lung cancers were detected at baseline; 6 out of 34 lung cancers were diagnosed after baseline but within 2 years follow-up. The median baseline volume of nodules subsequently diagnosed as lung cancers after baseline was 278.1 mm^3 (IQR 227.2–313.3 mm^3), significantly smaller than lung cancers diagnosed at baseline (1746.2 mm^3; IQR 1136.9–2849.8 mm^3) (p < 0.001). The median baseline nodule volume of all lung cancers combined was 1407.4 mm^3 (IQR 700.5–2533.1 mm^3). A total of 67.6% of lung cancers were located in the upper lobes and the same number were intraparenchymal (see Table 3 and Fig. 3). Most were found in stage I (62%), with adenocarcinoma (50%) and squamous cell carcinoma (32.4%) being the most frequent histological types.

3.4. Probability of lung cancer

The 2-year cancer probability for all indeterminate and positive solid or part-solid nodules was 9.1% (95% CI 6.6–12.5). For nodules with a volume of 100–300 mm^3, the 2-year lung cancer probability was 1.3% (95% CI 0.3–3.9), whereas for nodules of ≥300 mm^3 it was 22.1% (95% CI 16.0–29.8). Participants with nodules of ≥300 mm^3 had a significantly higher probability of developing lung cancer compared to those with nodules 100–300 mm^3 with relative risk (RR) of 18.1 (95% CI 5.6–58.4; p < 0.001). Solid or part-solid nodules located in the upper lobes had a 2-year cancer probability of 14.0% (95% CI: 9.5–20.2), and a significantly higher probability compared to other lobes with RR of 2.7 (95% CI 1.3–5.5; p = 0.009). The probability of cancer based on attachment (intraparenchymal vs non-intraparenchymal) and density (solid vs part-solid) categories showed no difference (see Table 4). The number of part-solid nodules and associated cancer events was limited, which constrained inference for this group. For non-solid nodules, no lung cancer events were observed; therefore, analyses for this subgroup were descriptive only and no inferential comparisons were performed.

4. Discussion

In this study, we performed a post hoc analysis using lung nodule data from the UKLS trial. Our main objective was to evaluate the distribution and characteristics of lung nodules detected on baseline CT scans and observe how many of these nodules developed into lung cancer within a two-year follow-up period. The distribution of indeterminate and positive lung nodules in a real-world screening setting has been rarely reported and visualized systematically [12,15]; this study seeks to address that gap. We found that solid and part-solid nodules with volume ≥300 mm^3 showed a significant predilection for the upper lobes compared to those measuring 100–300 mm^3 (OR 1.97; 95% CI 1.12–3.48). In contrast, nodules ≥300 mm^3 were significantly less likely to be intraparenchymal, although intraparenchymal location remained the most common across both size groups. No significant difference in nodule density was observed between the two volume categories. Among all solid and part-solid nodules detected, 9.1% were diagnosed as lung cancer within two years and of these, 61.8% were ≥300 mm^3 in volume and located in the upper lobes. A better understanding of the anatomical distribution and characteristics of lung nodules from

Table 2
Solid and part-solid lung nodule characteristics overall and by size categories; indeterminate (100–300 mm^3) and positive (≥300 mm^3).

Nodule characteristics	All solid and part-solid nodules	Solid and part-solid nodules	Solid and part-solid nodules	p value
	n = 373 (100%)	**100–300 mm^3 n = 233 (62.5%)**	**(≥ 300 mm^3) n = 140 (37.5%)**	
Volume (mm^3), median (IQR)	211.1 (128.3–474.8)	145.9 (114.6–194.5)	709.7 (422.0–1669.2)	
Location – lobe, n (%)				
Upper lobes	**164 (44.0)**	**92 (39.5)**	**72 (51.4)**	**<0.05*** [OR 1.97 (95% CI 1.12–3.48)]
RUL	97 (26.0)	61 (26.2)	36 (25.7)	
LUL	67 (18.0)	31 (13.3)	36 (25.7)	
Other locations	**209 (56.0)**	**141 (60.5)**	**68 (48.6)**	
RML	35 (9.4)	25 (10.7)	10 (7.1)	
RLL	89 (23.9)	59 (25.3)	30 (21.4)	
LLL	84 (22.5)	56 (24.0)	28 (20.0)	
Trachea	1	1	0	
Bifurcation	0	0	0	
Location/attachment type, n (%)				
Intraparenchymal	**244 (65.4)**	**169 (72.5)**	**75 (53.6)**	**<0.05**** [OR 0.43 (95% CI 0.25–0.74)]
Non-intraparenchymal	**114 (30.6)**	**58 (24.9)**	**56 (40.0)**	
Juxtapleural	24 (6.4)	14 (6.0)	10 (7.1)	
Pleural-based	90 (24.2)	44 (18.9)	46 (32.9)	
Unknown	15 (4.0)	6 (2.5)	9 (6.4)	
Density type, n (%)				
Solid	362 (97.1)	228 (97.9)	134 (95.7)	0.432***
Part solid	11 (2.9)	5 (2.1)	6 (4.3)	

IQR = inter quartile range; RUL = right upper lobe; RML = right middle lobe; RLL = right lower lobe; LUL = left upper lobe; LLL = left lower lobe.
* Nodules sized ≥300 mm^3 were significantly more likely to be located in upper lobes compared to nodules sized 100–300 mm^3.
** Nodules sized ≥300 mm^3 were significantly less likely to be intraparenchymal compared to nodules sized 100–300 mm^3.
*** There was no difference in density type between nodules sized ≥300 mm^3 and 100–300 mm^3.

screening program may support the development of standardized, population-based screening guidelines and help refine risk assessment beyond size alone.

All lung cancers appeared as solid nodules or part solid nodules and no cancers developed from nonsolid nodules. The two-year lung cancer probability was significantly higher in nodules ≥300 mm^3 compared to nodules of 100–300 mm^3 with RR of 18.1 (95% CI 5.6–58.4; $p < 0.001$). These findings highlight the importance of nodule volume ≥300 mm^3 as a potentially relevant marker for lung cancer risk in screen detected nodules, while nodules of 100–300 mm^3 require careful evaluation. Similarly, upper lobe nodules presented a significantly higher cancer probability compared to other lobe nodules with RR of 2.7 (95% CI 1.3–5.5; $p = 0.009$). However, it is worth noting that 51.4% of solid and part-solid nodules with a volume ≥300 mm^3 were located in the upper lobes. This overlap suggests that the observed association with lobe location may be partly driven by the distribution of larger nodules in the upper lobes and indicates potential collinearity between nodule volume and upper-lobe location, which limits the ability to fully separate their independent effects. Nevertheless, upper-lobe location appears to be associated with increased malignancy risk, consistent with the Pan-Canadian Early Detection of Lung Cancer Study (PanCan) findings which reported more cancers arising in the left and right upper lobes compared to the lower lobes or right middle lobe [16].

These results are similar to findings from the NELSON trial. Horeweg et al reported that NELSON participants with nodules ≥300 mm^3 had significantly higher lung cancer probabilities compared to participants without nodules. Both, the NELSON trial and our study, highlighted nodule volume as an important predictor for lung cancer risk. Horeweg et al also compared volume-based (100 mm^3 and 300 mm^3) and diameter-based (5 mm and 10 mm) cutoff for classifying indeterminate and positive nodules. They concluded that volume-based measurements offered higher specificity and positive predictive value, with similar sensitivity and negative predictive value [10,17]. These findings highlight the importance of selecting measurement approaches that are technically appropriate for different nodule subtypes. In this study, volumetric assessment was applied to solid and part-solid nodules, for which reliable segmentation is feasible, whereas diameter-based measurement was used for non-solid nodules because their indistinct margins and variable density limit reliability of volumetric assessment, remain consistent with current guidelines. Recent Lung-RADS (2022), NELSON trial, and BTS guidelines preferentially recommend volumetric measurements where reliable, while retaining diameter-based criteria when volumetry is not feasible or appropriate [8,9]. Overestimating nodule size may lead to unnecessary diagnostic tests and in contrast, underestimating nodule size could result in delayed diagnosis due to false-negative outcomes [13,18]. Therefore, as our study indicated volume to be important predictor of malignancy, this emphasizes the importance of accurate nodule measurement in order to determine the most appropriate nodule management strategy. Further, volume measurement provides a three-dimensional representation of nodule size. This approach is particularly useful for capturing the true size of irregularly shaped nodules, offering a single and consistent metric that reflects overall nodule extent [19].

Only a few lung cancer screening trials have reported detailed nodule characteristics and their association with subsequent lung cancer development. One report from NELSON trial described nodule locations and characteristics from three screening rounds over three years and resulted in 40.6% (200/493) of participants with positive screening results or noncalcified nodules with volume greater than 500 mm^3 were diagnosed with lung cancer [12]. With a different screening program setting compared to NELSON, including a single-round design and different inclusion criteria, our study showed a lung cancer rate of 10.5% (34/324) among participants with solid or part-solid nodules with volume ≥100 mm^3 and 25.4% (31/122) among those with nodules ≥300 mm^3 at baseline screening, after a two-year period. Our study and NELSON identified the right upper lobe as having the highest percentage of cancers (36.3% vs 45.0%, respectively), but our study found also a notably high percentage in the left upper lobe (32.3%) which was different compared to NELSON (19.1%) [12]. These upper lobe findings align with results from other studies such as the NLST, PanCan, and in

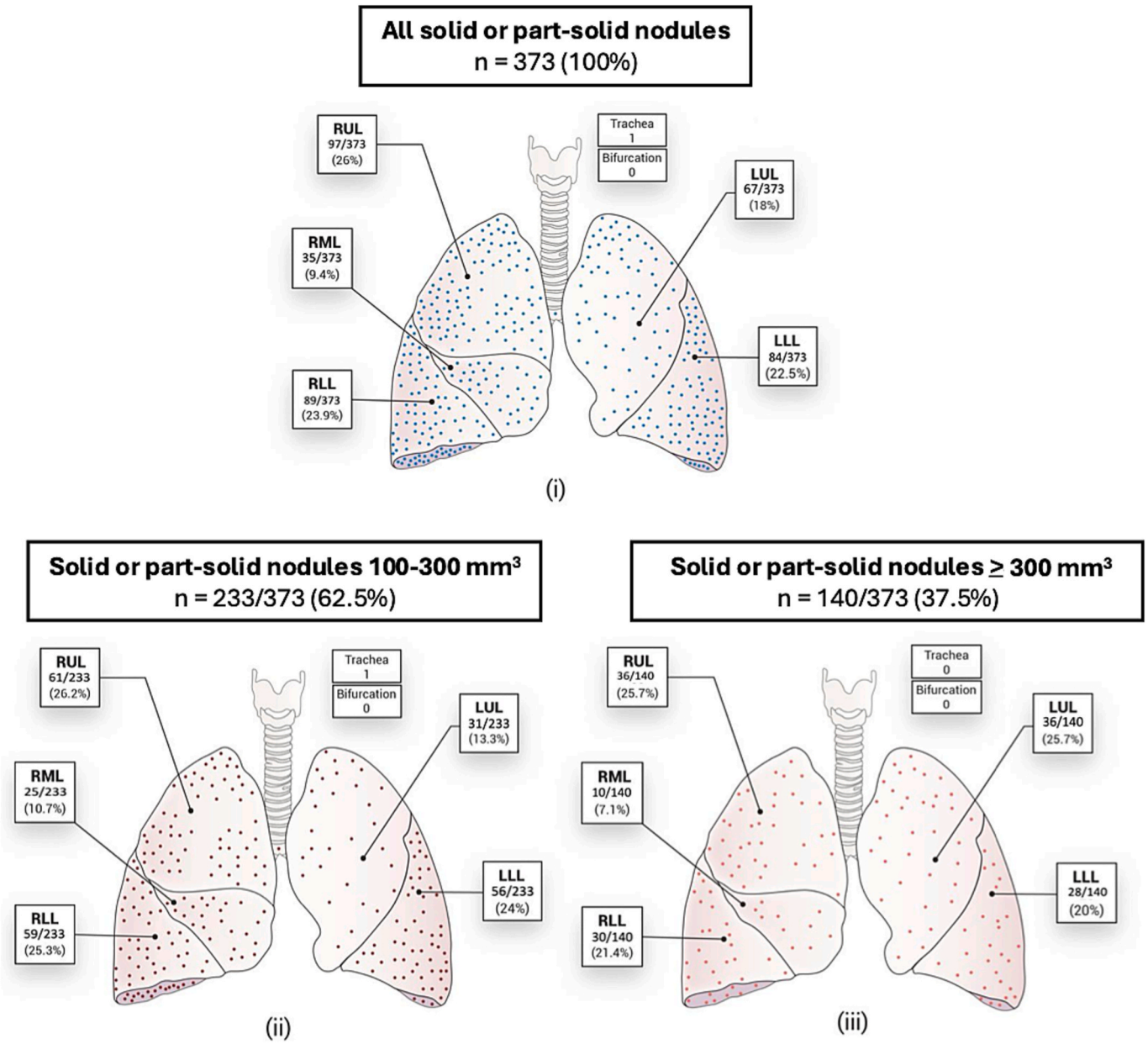


Fig. 2. Location of all nodules: (i) All solid or part-solid nodules; (ii) All solid or part-solid nodules 100–300 mm^3; (iii) All solid or part-solid nodules ≥300 mm^3 RUL = right upper lobe; RML = right middle lobe; RLL = right lower lobe; LUL = left upper lobe; LLL = left lower lobe.

Asian populations [4,16,20]. Biological hypotheses have been proposed that upper lobes may accumulate higher concentrations of inhaled carcinogens due to differences in ventilation, perfusion, slower lymphatic drainage, and clearance mechanisms, potentially increasing cancer risk [21,22]. However, findings related to upper-lobe location in the present study should be interpreted in the context of nodule volume, as larger nodules were more frequently located in the upper lobes and multivariable adjustment was not feasible.

Our study observed a two-year lung cancer probability of 9.1% (95% CI 6.6–12.5) from all radiologically suspicious solid and part-solid nodules. The NELSON trial reported a lower probability of 2.5% (95% CI 2.1–2.9) from all participants with CT-detected nodules over two years and two screening rounds. This difference may be explained by the fact that our probability was calculated only from radiologically suspicious solid or part-solid nodules. In our study, the volume cutoff for defining suspicious nodules was ≥100 mm^3, while NELSON used a broader range of ≥50 mm^3. The use of $LLP_{version}2$ for selecting high-risk individuals likely contributed the higher lung cancer probability in our study, supported by significantly higher $LLP_{version}2$ scores in lung cancer participants ($p < 0.05$). For nodules ≥300 mm^3, both studies showed high probability of developing lung cancer with 22.1% (95% CI 16.0–29.8) in our study and 16.9% (95% CI 14.1–20.0) in NELSON [10].

In this study we focused on baseline detected lung nodules, yet a lung cancer screening program is ongoing screening process where individuals undergo short-term and annual CT-scans to monitor lung nodule growth over time. A key constraint of this study is the restriction to single-round baseline analysis without incorporation of longitudinal nodule growth information. Therefore, a limitation of this study is the absence of nodule growth data, which are typically assessed using volume doubling time (VDT) to determine the appropriate management strategy for lung nodules. VDT is important for assessing growth, especially in nodules below 300 mm^3 where longer follow-up may be required [23]. In addition, although justified by adherence to guidelines, the differences in measurement approaches across nodule types

Table 3
Overview of lung cancer participant characteristics, stratified by diagnosis at baseline and within a 2-year follow-up period.

Variables	All lung cancer cases n = 34	Diagnosed at baseline n = 28 (82.4%)	Diagnosed after baseline within 2 years **n = 6** (17.6%)	p value
Volume (mm^3), median (IQR)	1407.4 (700.5–2533.1)	1746.2 (1136.9–2849.4)	278.1 (227.2–313.3)	**<0.05***
Volume 100–300 mm^3	3 (8.8)	0	3	
Volume ≥300 mm^3	31 (91.2)	28	3	
Location – lobe, n (%)				
Upper lobes	**23 (67.6)**	**18**	**5**	
RUL	12 (35.3)	9	3	
LUL	11 (32.3)	9	2	
Other lobes	**11 (32.4)**	**10**	**1**	
RML	3 (8.8)	2	1	
RLL	4 (11.8)	4	0	
LLL	4 (11.8)	4	0	
Location/ attachment type, n (%)				
Intraparenchymal	23 (67.6)	19	4	
Non-intraparenchymal	8 (23.6)	6	2	
Juxtapleural	1 (2.9)	1	0	
Pleural-based	7 (20.6)	5	2	
Unknown attachment	3 (8.8)	3	0	
Density type, n (%)				
Solid	31 (91.2)	26	5	
Part-solid	3 (8.8)	2	1	
Stage (TNM version 8 staging), n (%)				
I	21 (61.7)	16	5	
II	7 (20.6)	7	0	
III	4 (11.8)	4	0	
IV	2 (5.9)	1	1	
Histology, n (%)				
Adenocarcinoma	17 (50.0)	12	5	
Mucinous adenocarcinoma	1 (2.9)	1	0	
Squamous cell carcinoma	11 (32.4)	10	1	
Small cell carcinoma	2 (5.9)	2	0	
Acinar cell carcinoma	1 (2.9)	1	0	
Bronchogenic carcinoma	1 (2.9)	1	0	
Metastasis of small cell carcinoma	1 (2.9)	1	0	
Treatment, n (%)				
Surgery alone	19 (55.9)	15	4	
Surgery + chemotherapy	8 (23.5)	8	0	
Surgery + radiotherapy	1 (2.9)	1	0	
Radiotherapy alone	1 (2.9)	1	0	
Palliative**	3 (8.8)	2	1	
Unknown	1 (2.9)	1	1	

IQR = inter quartile range; RUL = right upper lobe; RML = right middle lobe; RLL = right lower lobe; LUL = left upper lobe; LLL = left lower lobe.
*The median volume of lung cancer diagnosed at baseline was significantly larger than those diagnosed after baseline within 2 years.
** Palliative treatment included best supportive care (n = 2; stage III and IV) and palliative multimodal therapy consisting of surgery, radiotherapy for metastases, and chemotherapy (n = 1; stage IV).
All data are presented at the participant-level (n = 34); nodule characteristics refer to the corresponding nodule associated with each lung cancer case.

introduced methodological heterogeneity. Although not feasible in this study, future research should investigate distribution and characteristics of lung nodules at follow-up CT screening, as once these nodules are deemed to be growing and therefore suspicious for lung cancer, standardized management strategies are required to streamline diagnostic work-up in the clinic. Further, the limited ethnic diversity in the UKLS study may also affect generalizability of our results and should therefore be interpreted with caution when applied to different ethnic populations. We also did not analyze nodules smaller than 100 mm^3. These cases would typically be classified as a negative result and the individual would proceed to the next screening round. Lastly, we were only able to estimate the two-year probability of lung cancer for each nodule characteristic individually, without incorporating combinations of multiple predictors into a single model. This limitation was primarily due to convergence issues encountered during model fitting, likely related to limited number of lung cancer events, data sparsity across certain variable combinations, and the complexity introduced by clustering within participants. As a result, we are unable to draw robust conclusions about the independent contributions or potential interactions of individual risk factors such as smoking status or lung comorbidities when considered together.

As lung cancer screening implementation progresses, robust, standardized nodule management guidelines are necessary to aid clinicians once a nodule has been detected which has an increased risk of being lung cancer. To develop such guidelines for follow-up, immediate referral, and diagnostic work-up, evidence, such as that presented in this study, is needed from different lung cancer screening populations worldwide.

4.1. Conclusion

In this post hoc analysis of UKLS trial data, we found that solid and part-solid nodules volume ≥300 mm^3 were more commonly located in the upper lobes and less frequently intraparenchymal than nodules volume 100–300 mm^3, although intraparenchymal location remained the most prevalent overall. Most lung cancers diagnosed within two years were located intraparenchymally and in the upper lobes and were stage I adenocarcinomas or squamous cell carcinomas. Solid and part-solid nodules ≥300 mm^3 made up 37.5% of nodules but accounted for 91.2% of lung cancers within two years, highlighting volume ≥300 mm^3 as a potentially relevant marker for lung cancer risk. These findings highlight the importance of lung nodule volume and anatomical location

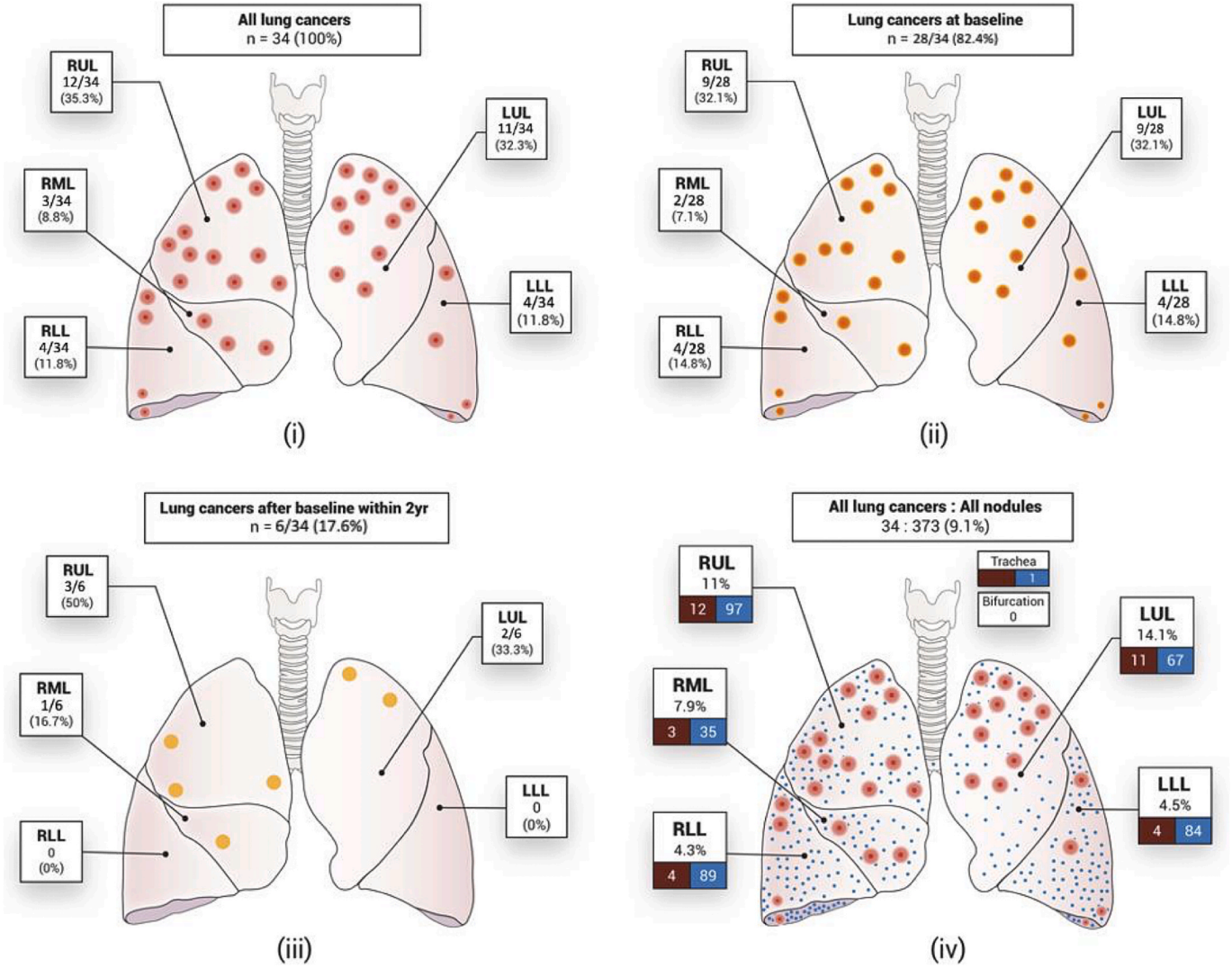


Fig. 3. Location of lung cancers: (i) All lung cancer cases; (ii) Lung cancers diagnosed at baseline; (iii) Lung cancers subsequently diagnosed as lung cancers after baseline within two years; (iv) Overlap of all lung cancers and lung nodules. RUL = right upper lobe; RML = right middle lobe; RLL = right lower lobe; LUL = left upper lobe; LLL = left lower lobe. Percentages in figure (iv) represent the proportion of lung cancers (brown boxes) relative to the total number of radiologically suspicious solid and part-solid nodules (blue boxes) within each lobe. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

in lung cancer risk prediction. Future research should focus on distribution and characteristics of lung nodules detected at follow-up screening and should include larger and more diverse populations.

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CRediT authorship contribution statement

Oke Dimas Asmara: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Harriet L. Lancaster:** Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Data curation, Conceptualization. **John K. Field:** Writing – review & editing, Validation, Supervision, Resources, Data curation. **Michael P.A. Davies:** Writing – review & editing, Validation, Supervision, Resources, Data curation. **Stephen W. Duffy:** Writing – review & editing, Supervision. **E. Christiaan Boerma:** Writing – review & editing, Supervision. **Edith Visser:** Writing – review & editing, Methodology, Formal analysis. **Kim de Jong:** Writing – review & editing, Methodology, Formal analysis. **Anand Devaraj:** Writing – review & editing, Supervision, Resources, Data curation. **Marjolein A. Heuvelmans:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Data curation, Conceptualization. **Wouter H. van Geffen:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Data curation, Conceptualization. **Matthijs Oudkerk:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Investigation, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interest: Michael P.A. Davies reports receiving support from the Roy Castle Lung Cancer Foundation. John K. Field reports receiving grants/contracts from EU Horizon, MRC-Saver, CRUK, IAA, Elypta, and Therapeutic

Table 4
Nodule-level probability of lung cancer based on baseline lung nodule characteristics.

No	Characteristics	Probability with 95% CI	
1.	**All solid or part-solid nodules**	34/373	9.1% (95% CI 6.6–12.5)
2.	**Volume**		
	Solid or part-solid nodules ≥300 mm^3	31/140	22.1% (95% CI 16.0–29.8)
	Solid or part-solid nodules 100–300 mm^3	3/233	1.3% (95% CI 0.3–3.9)
	≥ 300 mm^3 vs 100–300 mm^3		RR = 18.1 (95% CI 5.6–58.4) p < 0.001**
3.	**Lobe**		
	Located in upper lobes	23/164	14.0% (95% CI 9.5–20.2)
	Located in other lobes	11/209	5.3% (95% CI 2.9–9.3)
	Upper vs other lobes		RR = 2.7 (95% CI 1.3–5.5) p = 0.009**
4.	**Attachment***		
	Intraparenchymal	23/244	9.4% (95% CI 6.3–13.8)
	Non-intraparenchymal	8/114	7.0% (95% CI 3.4–13.4)
	Intraparenchymal vs non-intraparenchymal		RR = 1.3 (95% CI 0.4–5.0) p = 0.497**
5.	**Density**		
	Solid nodules	31/362	8.5% (95% CI 6.1–11.9)
	Part-solid nodules	3/11	27.3% (95% CI 9.2–57.1)
	Solid vs part-solid		RR = 2.6 (95% CI 0.8–8.1) p = 0.107**

*Attachment type was unavailable for 15 lung nodules.
**RR from mixed-effects logistic regression models with random intercept.

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Appendix A. Supplementary data

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